

Review

The dynein microtubule motor

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*Department of Biochemistry, University of Connecticut Health Center, 263 Farmington Avenue, Farmington, CT 06032-3305, USA***Abstract**

Dyneins are large multi-component microtubule-based molecular motors involved in many fundamental cellular processes including vesicular transport, mitosis and ciliary/flagellar beating. In order to achieve useful work, these enzymes must contain motor, cargo-binding and regulatory components. The ATPase and microtubule motor domains are located within the very large dynein heavy chains that form the globular heads and stems of the complex. Cargo-binding activity involves the intermediate chains and several classes of light chain that associate in a subcomplex at the base of the soluble dynein particle. Regulatory control of dynein motor function is thought to involve the phosphorylation of various components as well as a series of light chain proteins that are directly associated with the heavy chains. These latter polypeptides have a variety of intriguing attributes, including redox-sensitive vicinal dithiols and Ca^{2+} -binding, suggesting that the activity of individual dyneins may be subject to multiple regulatory inputs. Recent molecular, genetic and structural studies have revealed insight into the roles played by these various components and the mechanisms of dynein-based motility. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

Dyneins are massive molecular motor complexes of 1–2 MDa that generate force towards the minus end of microtubules. The coordinated action of flagellar dyneins, which comprise the inner and outer arms of the eukaryotic axoneme, generates specific flagellar waveforms and thus provides the motive force necessary to move the cell through the solution (reviewed in [1,2]). In contrast, cytoplasmic dyneins are involved in a variety of intracellular motile processes including mitosis, maintenance of the Golgi apparatus and the trafficking of membranous vesicles

and other intracellular particles [3]. This divergence in function has given rise to the notion that flagellar and cytoplasmic dyneins are quite distinct entities. This idea has been further supported by the identification of multiple dynein heavy chain (HC) isoforms within a single organism. For example, the biflagellate green alga *Chlamydomonas* is now known to contain at least 16 different dynein heavy chains [4], at least 14 of which are thought to function in the flagellum. However, recent analysis of the accessory proteins associated with the dyneins from various sources argues against this view of the dynein family. Rather, these studies demonstrate strong similarities between cytoplasmic dynein and several flagellar isoforms and suggest that the major dichotomy in dynein form exists between those enzymes which contain multiple motor units and those with only a single ATP-sensitive microtubule-interacting domain.

Abbreviations: HC, heavy chain; IC, intermediate chain; LC, light chain; nNOS, neuronal nitric oxide synthase

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2. The motor units

Dynein motor activity is due to the action of the ~ 530 kDa HCs. The overall architecture of the dynein particle and the location of specific functional sites within the HCs are illustrated in Fig. 1a,b. In vitro assays have demonstrated that the rate of dynein-driven microtubule motility is considerably greater than that observed for kinesin and can reach values $> 6 \mu\text{m s}^{-1}$ [5–7]. These experiments have also revealed that some (but apparently not all) dyneins impart torque during the mechanochemical cycle and thus, in the in vitro gliding assay, cause rotation of the microtubule being translocated [8,9]. Cytoplasmic dyneins appear to be homodimers of HCs with presumably identical motor properties. However, some flagellar dyneins contain multiple heavy chains with different intrinsic motor activities which raises intriguing questions about how these can combine to yield a coordinated power stroke [10–12]. This is especially enigmatic as genetic dissection of outer arm dynein combined with image averaging of electron micrographs has revealed that each individual head potentially has continual direct access to the adjacent doublet microtubule (Fig. 2a) [13,14]. Nanometer scale tracking has indicated that individual dynein motors are capable of moving microtubules in vitro suggesting that dynein is a processive motor. Unlike kinesin, however, dynein does not track for long distances along single protofilaments but rather moves laterally over the microtubule surface during translocation [15].

Each HC comprises an N-terminal domain of ~ 160 kDa which forms the stem of the structure and interacts with additional HCs and other components (Fig. 1b). In vitro binding studies have identified an ~ 200 residue segment within this region that can mediate both HC–HC dimerization and HC–IC interactions [16]. Much of the specificity in terms of which components are present in a particular complex presumably must be encoded in these regions (see below). The C-terminal segments of the HCs form the ~ 350 kDa globular head domains (Fig. 2b). This includes a highly conserved central section of ~ 100 kDa that contains four nucleotide binding motifs (P1–P4). The P1 motif marks the ATP hydrolytic site and indeed ablation of this site renders the HC enzymatically inactive. The role played by the

other nucleotide binding motifs is unclear. However, phase partition and fluorescence anisotropy measurements using sea urchin sperm dynein have provided convincing evidence that at least two and perhaps all four sites do indeed bind ATP with physiologically significant binding constants of 10^4 – 10^5 M^{-1} [17,18].

The microtubule-binding domain is thought to be located at the tip of a slender stalk emanating from the head [19,20]. This structure apparently is formed from two coiled coil segments arranged in an anti-parallel manner that are encoded C-terminal to the P4 loop. The actual microtubule-binding site appears to be a small globular unit located between the coiled coil stretches. This region is thought to correspond to the stalk-like structures that have been observed protruding from the heads of individual dynein HCs prepared for electron microscopy by the quick freeze/deep etch procedure [21]. Three-dimensional reconstructions of the dynein motor domain from high-resolution negative stain images reveal a 13.5 nm diameter spheroidal structure composed of seven globular subdomains arranged around a central cavity [22] (Fig. 2c). These observations have intriguing implications as they suggest that the structural mechanism by which dynein generates force may be fundamentally different to kinesin and myosin [19,20]. For example, with this arrangement of structural elements, there are potentially two hinge locations at which ATP-induced alterations in conformation could occur during the mechanochemical cycle. The first hinge could be located between the globular head unit and the microtubule-binding stalk, and the second between the head and the N-terminal stem region that forms the cargo-binding tail of the enzyme.

3. Regulating motor function

Dynein motor function is under exquisite and precise control both in the cytoplasm and the flagellum. For example, flagellar dyneins must be activated sequentially to allow a wave of activity to be propagated from the base to the tip. To obtain a flagellar bend, the arms in one hemisphere of the cylindrical axoneme must be active, while those in the opposite hemisphere are deactivated. Furthermore, the region of active sliding must be switched from one hemi-

sphere to the other in order to enable both principal and reverse bends to be formed (see [23] for recent review). Finally, both these activities must be switched at a rate consistent with the beat frequency of the flagellum: in wild-type *Chlamydomonas* this value is ~ 60 Hz. Likewise, the action of dyneins in the cytoplasm also must be tightly controlled to allow for the temporal and spatial regulation necessary to enable a particular cargo to be transported to the appropriate place at the correct time. For example, during mitosis the activity of cortically anchored dyneins on astral microtubules must be controlled so that they contribute to spindle pole separation at the appropriate time during anaphase.

How might dynein-based motor activity be controlled? A number of possibilities for which there is some evidence exist including the following.

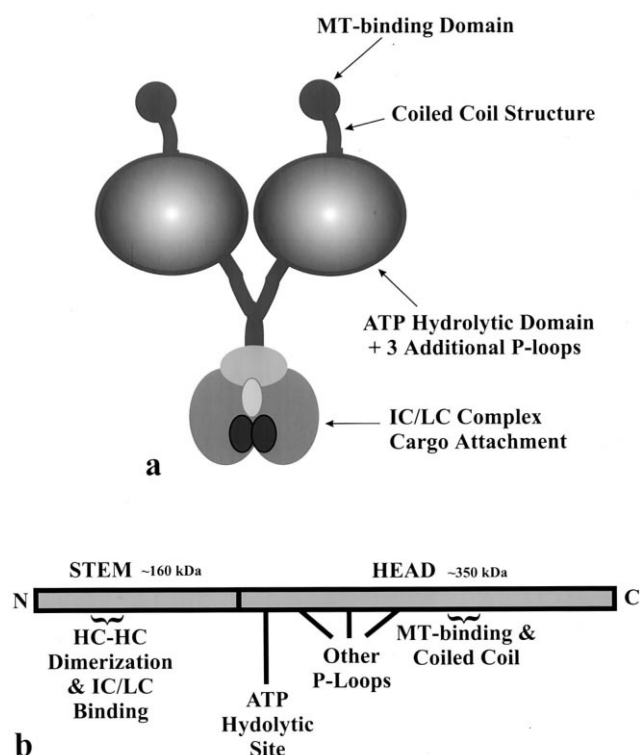


Fig. 1. Dynein architecture and heavy chain organization. (a) Model of a generic two-headed dynein particle indicating the general location of various structural and functional domains described in the text. (b) Map of the dynein heavy chain identifying the segments of these large proteins involved in binding the IC/LC complex, ATP and microtubules.

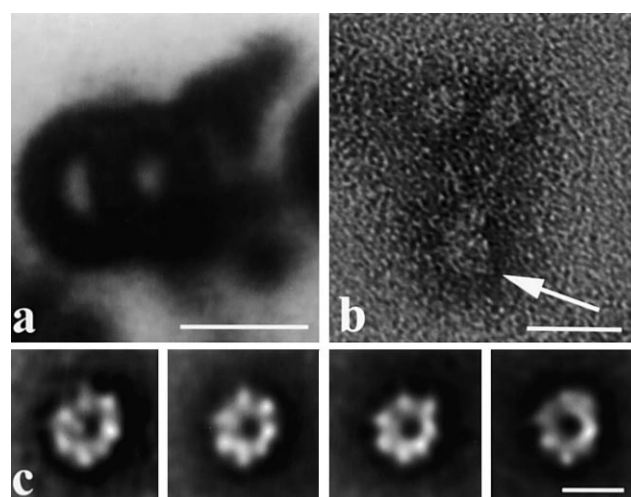


Fig. 2. Structure of the dynein motor. (a) Inner and outer dynein arms associated with flagellar doublet microtubules from *Chlamydomonas*. This averaged image from 16 axonemal cross sections was provided by Dr. Ritsu Kamiya (University of Tokyo). The three lobes (from outermost to innermost) of the outer arm with access to the adjacent doublet are formed from the motor domains of the α , β and γ HCs, respectively. Bar = 25 nm. (b) Negative stain electron micrograph of a two-headed portion of the *Chlamydomonas* outer dynein arm containing the α and β HCs and the IC/LC complex. This particular preparation had been incubated with a monoclonal antibody raised against the IC2 component. The antibody is bound at the base of the particle (arrow) indicating the location of IC2. Bar = 25 nm. From [52]. (c) Averaged images of the negatively stained *Dictyostelium* cytoplasmic dynein motor domain. The 13.5-nm diameter globular head is formed from seven distinct lobes surrounding a central cavity. This gallery was provided by Dr. Michael Koonce (Wadsworth Center). Bar = 10 nm. From [22].

3.1. Direct phosphorylation/dephosphorylation of the HCs

This is known to occur *in vivo* for certain inner and outer arm HCs [24,25], and *in vitro* experiments using isolated axonemes strongly suggest that the kinases and phosphatases involved are integral stable components of the microtubular axoneme [24]. Mapping experiments indicate that major sites of phosphorylation within the α HC of *Chlamydomonas* outer arm dynein occur within ~ 20 kDa of the P1 ATP binding loop and at or near the microtubule-binding stalk [24]. *In vivo* labeling of cytoplasmic dynein in rat optic nerve has demonstrated that the HC is differentially phosphorylated depending on whether the dynein derived from the total cellular pool or from

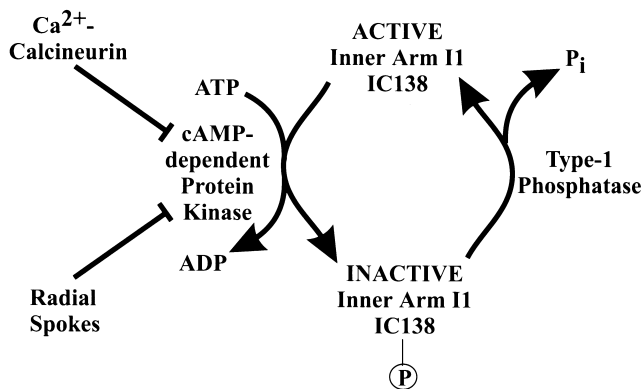


Fig. 3. Cyclic AMP-dependent control of inner arm II function. The model proposed by Habermacher and Sale [32] for the control of inner arm II activity through the cyclic modification of IC138 by cAMP-dependent protein kinase and subsequent dephosphorylation by a type-1 phosphatase. The radial spokes negatively impinge on this system to control kinase activity presumably through the dynein regulatory complex. It has been further proposed by King and Dutcher [34] that Ca^{2+} also affects the relative activities of the enzymes responsible for modifying IC138 and that this allows for a Ca^{2+} -dependent control of flagellar beating and phototactic turning. This Ca^{2+} effect is mediated by the Ca^{2+} -dependent phosphatase calcineurin that dephosphorylates the regulatory subunit of cAMP-dependent kinase, allowing it to rebind the catalytic subunit with consequent downregulation of kinase activity.

vesicles moving in the anterograde direction (where dynein is presumably inactive). These results strongly support the hypothesis that direct HC phosphorylation can act as a switch for dynein-based motility [26].

3.2. Phosphorylation of ICs and LCs that result in an alteration in motor activity

The cAMP-dependent phosphorylation of outer arm dynein LCs from *Paramecium* cilia [27,28] and sea urchin sperm flagella [29] has been observed. Furthermore, modification of this 29 kDa *Paramecium* protein was correlated with a decrease in the velocity of microtubule translocation in an in vitro assay suggesting a direct effect on the motor properties of the enzyme.

In *Chlamydomonas*, isolation of suppressor mutants that could restore motility to strains paralyzed by defects in the radial spokes and central pair complex identified a dynein control pathway involving a distinct regulatory complex [30,31]. Subsequent ge-

netic and biochemical experiments have revealed that the phosphorylation/dephosphorylation of a 138 kDa IC associated with inner arm II determines the rate of microtubule sliding in disintegrating axonemes [32]. The cAMP-dependent modification of this component is controlled through the radial spoke system [33] and may play a role in phototactic turning [34]. Although the precise relationship between the dynein regulatory complex and the cAMP-dependent protein kinase and type-1 phosphatase responsible for modification of inner arm II is still uncertain, it is quite clear that the motor function of at least one flagellar inner arm dynein is controlled by this modification (Fig. 3). However, although suppressor mutations have been identified in both β and γ HCs [35,36], the mechanism by which the dynein regulatory complex impinges on the outer arms remains unknown.

3.3. Ligand-induced conformational changes in LCs directly associated with the HCs

A number of flagellar dynein HCs are directly associated with LCs that can bind Ca^{2+} , raising the possibility that these enzymes are controlled by direct ligand binding in a manner analogous to that used in adductor muscle myosin [37]. For example, a subset of *Chlamydomonas* inner dynein arms contain centrin, a Ca^{2+} -binding homologue of calmodulin [31], and the γ dynein HC from the outer arm interacts directly with another EF-hand protein (LC4) [38]. In *Chlamydomonas*, both phototaxis (differential regulation of beat frequency) and photoshock (an alteration in waveform leading to flagellar reversal) are Ca^{2+} -mediated. The former is caused by alterations in flagellar Ca^{2+} levels in the submicromolar range [39] and occurs in cells lacking outer arms suggesting that centrin might be involved in this process. In contrast, the photoshock response is caused by an increase in intraflagellar Ca^{2+} to 10^{-4} M (flagella stop beating at 10^{-5} M and undergo reversal at 10^{-4} M) and is aberrant in strains lacking outer arms [40–42]. The LC4 protein, which is thought to be the only Ca^{2+} -binding component of the outer arm, has a K_{Ca} of $\sim 10^{-5}$ M [38]. This affinity is appropriate for the LC4 protein to act as a Ca^{2+} sensor during the avoidance response. Finally, the idea that Ca^{2+} -controlled waveform conversion is

mediated directly by the dynein arms is supported by the observation that neither the radial spokes nor central pair microtubule structures are required [43].

3.4. Direct action of additional non-dynein components

It is conceivable that proteins not considered direct components of a dynein may be involved in motor regulation. This idea is especially attractive in the flagellum where the potential for spatially precise interactions exists. Recent analysis of LC1 from the *Chlamydomonas* outer arm has determined that this polypeptide associates directly with the motor unit (not the stem) of the γ HC [44]; the solution structure of this protein is currently under investigation [45]. Furthermore, LC1 interacts directly in situ with an additional axonemal component of ~ 45 kDa. Thus, this 45 kDa protein, which has yet to be identified, is intimately associated with the motor domain of the γ HC and could therefore play a regulatory role.

3.5. Control by redox poise?

Molecular cloning of dynein-associated proteins has revealed an unanticipated functional motif namely the dithiol-containing redox-active site of the thioredoxin family. As a class, these small proteins act in a redox-sensitive manner to alter the oxidation state of vicinal dithiols (i.e., those arranged such that oxidation leads to formation of a disulfide bond) on other proteins (Fig. 4a). In *Chlamydomonas*, the light chains associated with the stems of the α and β HCs are both thioredoxins containing vicinal dithiols (Fig. 4b,c) and these LCs exhibit the high affinity for trivalent metals such as arsenic that is characteristic of the thioredoxin family [46]. The sea urchin outer arm IC1 protein (120 kDa) has a modular construction with an N-terminal thioredoxin unit followed by three tandem repeats of nucleoside diphosphokinase and a C-terminal region [47]. Identification of these proteins has raised the intriguing possibility that redox poise plays a role in manipulating flagellar dynein motor activity perhaps in response to global changes in cytoplasmic levels of reducing equivalents. For example, the redox state

(and pH) of the *Chlamydomonas* cytoplasm varies considerably depending on whether the cells are in the light or dark. Intriguingly, a redox-regulated cAMP-dependent signaling system involving tyrosine phosphorylation has recently been proposed to control capacitation of human sperm [48].

Biochemical studies have indicated that sulfhydryl oxidation of sea urchin and *Tetrahymena* outer arm dynein leads to the activation of ATPase activity [49,50]. Importantly, activation of the inner dynein arms does not seem to occur by this mechanism. Analysis of *Chlamydomonas* mutant axonemes has confirmed the outer arm specificity of this reversible effect and further suggests that it is due to the specific activation of only one of the three HCs within that arm (A. Harrison, S.M. King, unpublished results). It will clearly be of great interest to determine whether the thioredoxin-like LCs and sulfhydryl-sensitive enzymatic properties are related.

Although there appear to be many systems by which dynein is regulated, it seems likely that these different regulatory mechanisms are not mutually exclusive. Indeed, one likely explanation for the astounding complexity of the dynein motor is that each enzyme can be regulated by several distinct mechanisms working on individual motor units within a single dynein particle.

4. The cargo-binding problem

In order to produce useful work, the dynein motor units must be attached to the appropriate cargo. This raises significant problems both in the flagellum where the motor must attach to a specific axonemal site and even more obviously in the cytoplasm where there are multiple distinct cargoes within a single cell type. What then are the cargo-binding components of the dynein complex and how can specific cargo selection be achieved?

Both cytoplasmic and outer arm dyneins contain two intermediate chains of ~ 70 – 80 kDa. Electron microscopic studies of both partially dissociated dynein [51] and dynein–antibody complexes [52] have demonstrated that these components are located at the base of the soluble dynein particle (see Fig. 2b). Molecular cloning has now revealed that these proteins are related as all are members of the WD-repeat

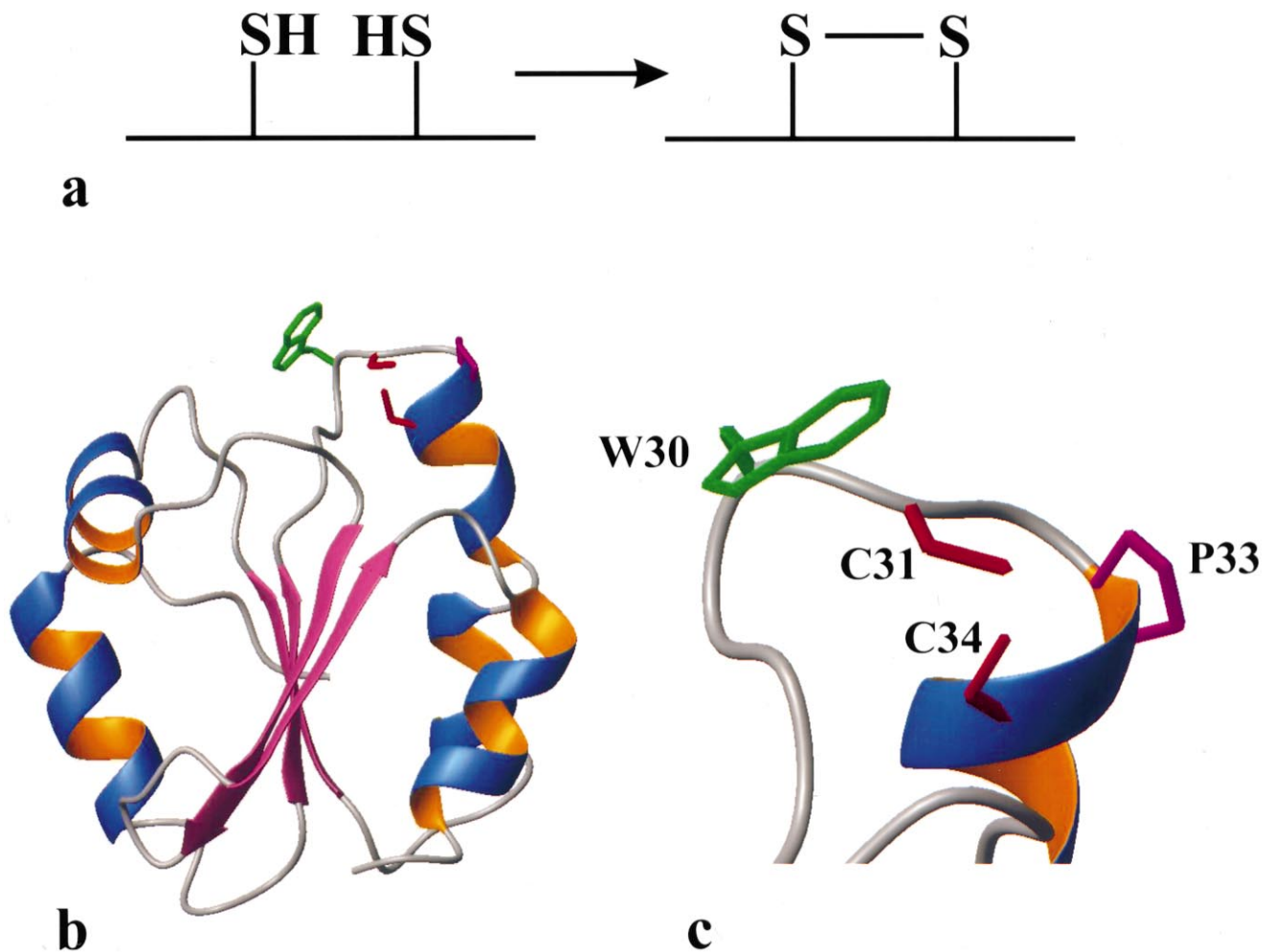


Fig. 4. Thioredoxin-like dynein light chain. (a) Both the α and β HCs in the *Chlamydomonas* outer arm are tightly associated with LCs related to thioredoxin [46]. Both LCs appear to be functional sulfhydryl oxidoreductases capable of oxidizing vicinal thiols to a disulfide. (b) Molecular model of LC4 associated with the α HC. This model was based on the *Escherichia coli* thioredoxin structure and was generated using SWISS-MODEL [115]. The LC is predicted to contain a central β sheet with α -helical segments stacked to either side. The WCGPC active site motif is located in a loop at the top of the model. (c) Magnified view of the redox-active site. Residue P33 (magenta) generates a sufficiently rigid bend in the backbone such that the side chains from the two C residues (red) are in close proximity and able to undergo oxidation. The completely conserved residue W30 is shown in green.

protein family and contain 6–7 repeat motifs in the C-terminal region of the polypeptide (Fig. 5a). Structurally, WD-repeat proteins are toroidal and composed of multiple 4-stranded twisted antiparallel β sheets that form the blades of the β -propeller [53]. Each WD motif contributes the outermost strand of one sheet and the three inner strands of the adjacent blade (Fig. 5b,c). This structural arrangement is stabilized by a hydrogen bond network (Fig. 5c) set up between the S, H, D structural triad and the W residue [53].

The N-terminal regions of the ICs vary consider-

ably in sequence and indeed it is these domains that appear to impart chain-specific functions. Both IC1 from the *Chlamydomonas* outer arm and IC74 of mammalian cytoplasmic dyneins apparently act directly in the attachment of the dynein motor to its cargo. In the former case, the IC interacts directly with tubulin and thus participates in structural attachment of the dynein arm to the flagellar microtubules [54,55]. However, this protein is not sufficient to attach the outer arm to its correct axonemal location. The precise attachment site is specified by a trimeric docking complex [56], although how the out-

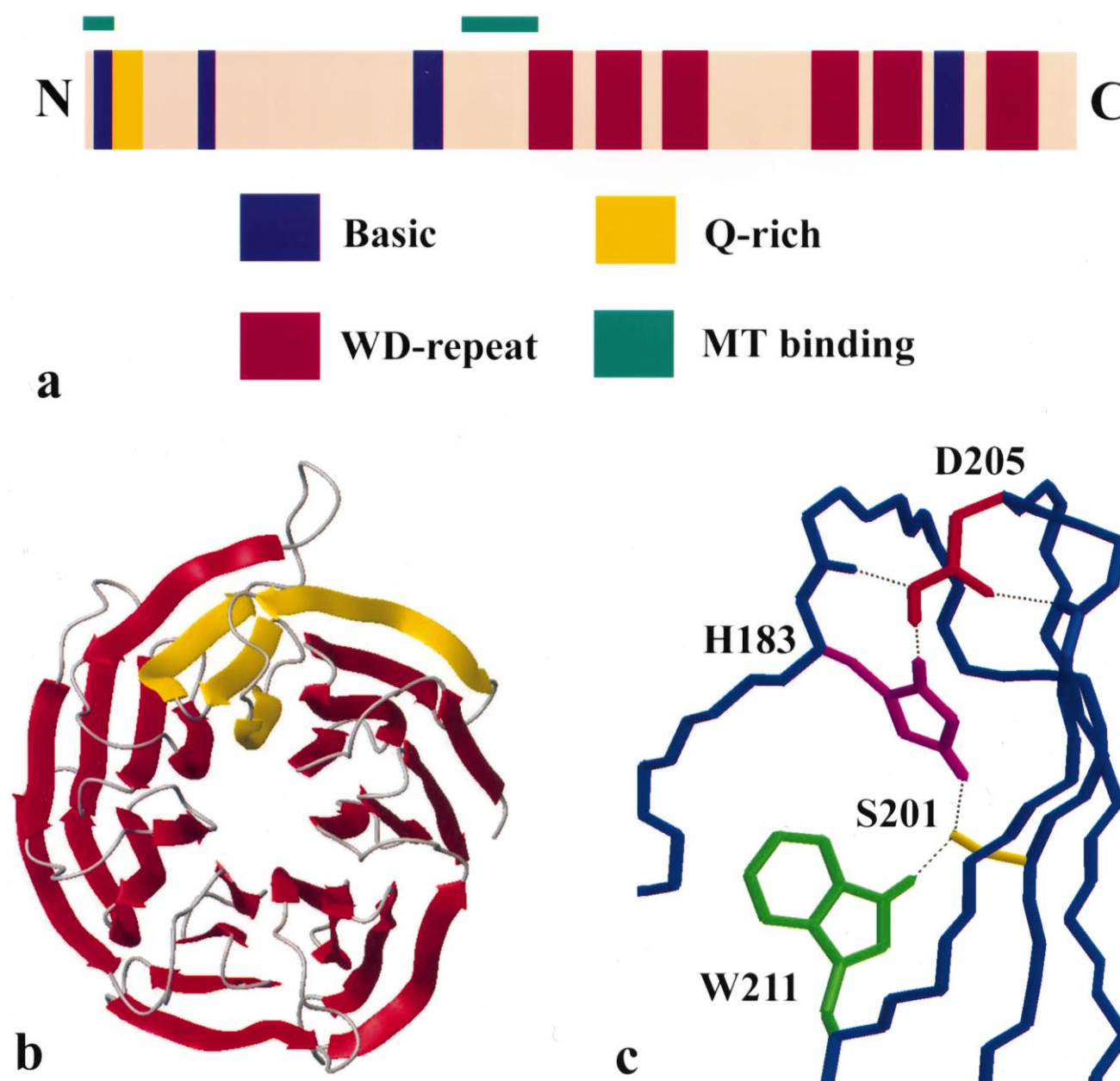


Fig. 5. Intermediate chains and structural features of WD-repeat proteins. (a) Map of IC1 from the *Chlamydomonas* outer dynein arm illustrating the C-terminal WD-repeat region that is predicted to form a β -propeller. It is this region which is relatively well conserved among ICs from various dyneins. The N-terminal segments are gene-specific; this domain of IC1 is involved in the stable attachment of outer arm dynein to the flagellar doublet microtubules. The major site of interaction with tubulin occurs through the N-terminal basic region [55]. (b) A ribbon diagram of the WD-repeat region from the G_β subunit of heterotrimeric G proteins is shown. The coordinates (pdb accession code 1gp2) are from [53]. This ~ 40 -residue repeat motif forms the outer strand of one antiparallel twisted β sheet and the inner three strands of the next sheet. One repeat (residues 171–212) is indicated in yellow. (c) Structural features of the WD repeat shown in yellow in b. The overall toroidal structure is stabilized by a hydrogen bond network. D205 (red) in the tight loop hydrogen bonds with two main chain amides and the side chain of H183 (magenta). The structure is further stabilized by hydrogen bonds involving H183, S201 (yellow) and W211 (green).

er arm interacts with this additional structure remains unclear. However, the mechanism of attachment is of great interest as it has consequences for flagellar beat frequency [57] and suggests that components at the base of the dynein particle play an important regulatory role. This idea is supported by the observation that mutant forms of IC2 in *Chlamydomonas* allow for outer arm assembly but do not result in wild-type motility [58].

Cytoplasmic dynein also contains two WD-repeat intermediate chains that are located at the base of the soluble particle [59,60]. However, unlike outer arm dynein, cytoplasmic dynein IC composition is complex and molecular and biochemical studies have revealed a plethora of isoforms derived from differential phosphorylation and alternative splicing [61]. These isoforms are differentially expressed during development of specific cell types such as neurons [62]. Although the functional significance of this isoform diversity remains uncertain, one obvious possibility is that different versions of IC74 are capable of interacting either with different proteins or with the same protein but with varying affinities.

What part does IC74 play in cytoplasmic dynein function? Its basal location strongly supports a role in mediating cargo attachment. In vitro binding studies have revealed that IC74 interacts directly with the p150^{Glued} component of dynactin [63]. This latter complex is an additional multimeric structure that acts as an activator of dynein-based motility [64]. Current models suggest that dynactin is bound at specific subcellular sites. Regulation of the IC74–dynactin interaction would then serve to attach the dynein motor to the particular cargo with dynactin acting essentially as an adapter.

Other dynein components also are located within the basal domain of the particle. Several of these, including the LC8 and Tctex1 LC proteins, are shared by both flagellar and cytoplasmic dyneins (the LC8, Tctex1 and *roadblock*/LC7 protein families are discussed in more detail below). Importantly, both LCs may mediate the attachment of specific cargoes to the dynein particle. For example, Tctex1 associates with the C-terminal tail of rhodopsin, and is involved in transport of the visual pigment to the base of the connecting cilium within the photoreceptor [65]. This protein also has been suggested to bind the Fyn tyrosine kinase in B lymphocytes [66], to

associate with and be phosphorylated by Fyn in *Torpedo* electric organ [67], and to interact with the Doc-2 protein where it may potentially mediate dynein attachment to specific vesicular structures [68]. Similarly, the LC8 protein, which is employed by several different enzyme systems (see below), has now been reported to mediate the specific recruitment of Bim (a member of the Bcl-2 family with pro-apoptotic activity) to dynein [69]. Disruption of this interaction by specific stimuli leads to cell death through the release of Bim and its subsequent ability to counteract the anti-apoptotic activity of Bcl-2.

As with motor activity, cargo attachment also needs to be tightly regulated to ensure that appropriate intracellular transport activities occur. Studies of dynein associations within *Xenopus* egg extracts have suggested that interaction of cytoplasmic dynein with vesicular structures may be controlled in a cell cycle-dependent manner by phosphorylation of the light intermediate chains. These 50–60 kDa dynein proteins contain an ATP-binding motif and are distantly related to ABC transporters [70,71]. However, there is as yet no direct evidence that they actually bind nucleotide and their precise location within the dynein particle has yet to be resolved.

5. Dynein LC8: an essential multi-enzyme component

The LC8 protein was first identified as an integral component of the *Chlamydomonas* outer dynein arm where it is associated with the intermediate chains at the base of the soluble particle [72,73]. More recently, molecular cloning of LC8 and examination of the EST database has revealed a group of very highly conserved proteins derived from mammals, nematodes and plants which share up to 93% sequence identity with the *Chlamydomonas* polypeptide [74]. Subsequent biochemical studies have determined that the LC8 protein is a stoichiometric component of both brain cytoplasmic dynein and myosin V [75,76]. Yeast two-hybrid screens have identified additional interactions involving the LC8 protein including a direct association with neuronal nitric oxide synthase (where it has been referred to as PIN) [77], I κ B α [78] and the Bcl-2 family protein Bim [69]. LC8 has also been identified in the tegument of Schistosomes [79].

The presence of the LC8 protein in many, seemingly unrelated, enzyme systems offers a simple explanation for the extremely high level of sequence conservation; namely that an alteration in LC8 structure would require simultaneous compensatory mutations in multiple separate enzymes. These observations have invited comparison of LC8 with calmodulin another highly conserved protein that interacts with numerous, otherwise unrelated, enzymes (e.g., [77,80]).

To date, LC8 mutants have been identified in four organisms. *Drosophila* partial loss-of-function alleles lead to morphogenetic defects in bristle and wing development, female sterility and also cause alterations in axonal guidance during development. Total loss-of-function results in embryonic lethality through the induction of apoptotic pathways [81, 82]. These phenotypes were originally interpreted as being caused by defects in cytoplasmic dynein; however, it now seems more likely that certain of these effects derive from the disruption of other enzyme activities. The *Aspergillus* LC8 homologue is encoded at the *NudG* locus. Temperature sensitive mutations lead to multiple dynein-mediated defects including the inhibition of nuclear migration through the failure to localize cytoplasmic dynein at the mycelial tip, reduced sporulation efficiency and slow growth [83].

In unicellular organisms, LC8 null mutations do not affect viability and indeed in *Saccharomyces cerevisiae* no phenotype has been observed [84]. However, in *Chlamydomonas* loss of LC8 (encoded at the *Fla14* locus) has profound consequences for motility as only very short flagella, lacking outer and inner dynein arms and radial spokes, are assembled [85]. Furthermore, lack of LC8 disrupts retrograde intra-flagellar transport (IFT) such that rafts of transported material accumulate in large clusters at the distal tips of the flagella. It remains unclear if the lack of axonemal dynein arms in the *fla14* mutant is due to the inability of LC8-minus dynein to assemble into the flagellum or whether such dyneins are simply not transported to the correct location to allow incorporation to occur. Analysis of additional *Chlamydomonas* mutants has indicated that retrograde IFT is mediated by the 1b isoform of cytoplasmic dynein [86]. Thus, it is quite possible that LC8 is required both by the IFT retrograde motor protein

itself and by several of the complexes that are being transported to the flagellar tip prior to assembly.

Biochemical analysis of cytoplasmic dynein [75] indicates that LC8 is present at a stoichiometry of two copies per complex whereas there appear to be four copies of this protein in *Chlamydomonas* outer arm dynein. Crosslinking experiments performed on the native proteins suggest that LC8 exists in situ as a dimer [80]. Studies using recombinant proteins in vitro also support the idea that LC8 has a tendency to oligomerize either to a dimer [87] or trimer [88]. However, the functional significance of this oligomerization activity remains unclear at present.

Recently, the solution structure of the rat LC8 (also known as PIN) protein has been solved by multi-dimensional NMR techniques [89]. The overall fold is shown in Fig. 6a,b. The N-terminal region forms a short β strand followed by adjacent α helices which form one surface. The C-terminal half of the protein comprises four antiparallel β strands. The structure is stabilized by an invariant W residue (W54) that protrudes from strand β 1 and undergoes an almost perfect base stacking arrangement with the side chain of F46 (Fig. 6c,d). W54 interacts with the aliphatic side chain of K87; a series of other nearby hydrophobic side chains including those from M13, M17, Y50 and L85 also contribute to the well-packed core. The interactions involving W54 result in the stabilization of an unusual *cis* bond between P52 and T53 in the tight loop connecting helix α 2 and strand β 1 and are thus essential for defining the overall fold.

LC8 presents two quite distinct surfaces to its environment. Acidic residues from helix α 1 and basic residues from helix α 2 (Fig. 7a) form a polar face. Molecular surface representations of this region (Fig. 7c–e) reveal two distinct patches of opposite charge. Interestingly, it is this region that is thought to be involved in LC8-LC8 interactions in situ as cross-linking with short amine-selective reagents requires helix α 2, suggesting that these regions are closely apposed [80]. The other face of LC8 is derived from side chains protruding from the β sheet. These form a hydrophobic concave surface with L84 at the base of the pocket (Fig. 7b,e). This hydrophobic region is ringed by hydrophilic residues, suggesting a structural similarity with calmodulin [89] where polar residues surrounding a hydrophobic patch are impor-

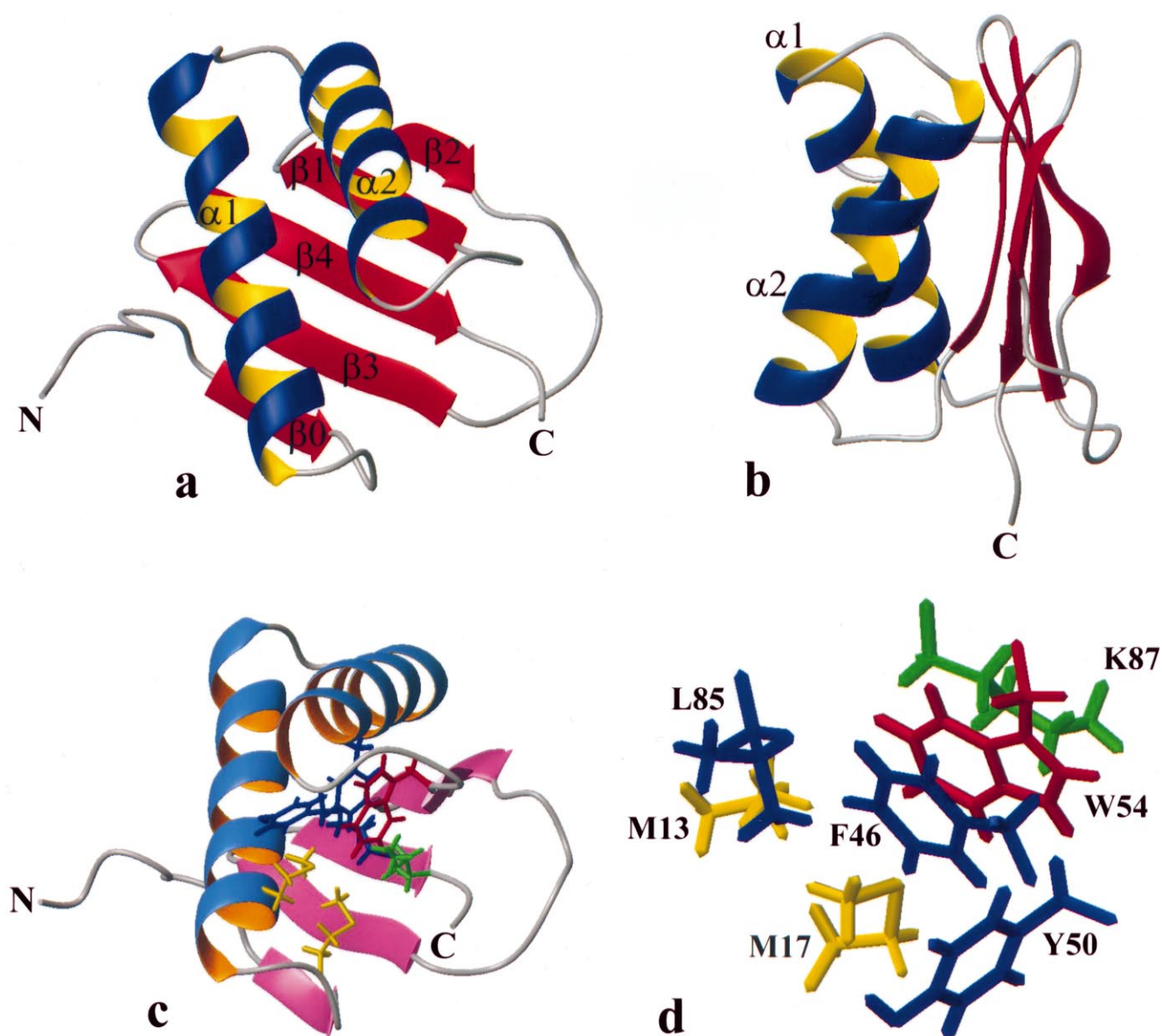


Fig. 6. NMR solution structure of LC8. Two views of the overall ribbon structure of LC8 are shown in panels a and b. The N-terminus forms a short β strand (β_0) followed by two antiparallel α helices. The C-terminal portion of the molecule contributes four antiparallel β strands. The very tight turn between helix α_2 and strand β_1 occurs because the peptide bond between P52 and T53 is forced to adopt the *cis* conformation. This unusual arrangement is stabilized by the almost perfect base stacking of the indole ring of W54 with the phenyl group of F46 (c,d). Other hydrophobic interactions involving side chains from M13, M17, Y50, L85 and the aliphatic portion of K87 also contribute significantly to the stability of the protein. The coordinates (pdb accession code 1bkq) are from [89].

tant in determining the binding specificity of the protein.

Although it is clearly essential for a variety of enzymes, the precise role played by LC8 in these systems is not yet certain. For example, the interaction of LC8 (PIN) with nNOS was originally de-

scribed as causing a dimer–monomer transition of nNOS thus leading to downregulation of enzyme activity [77]. However, recent reports suggest that nNOS activity is unaffected by LC8 binding [87]. The availability of a high-resolution structure combined with the null alleles now will allow for system-

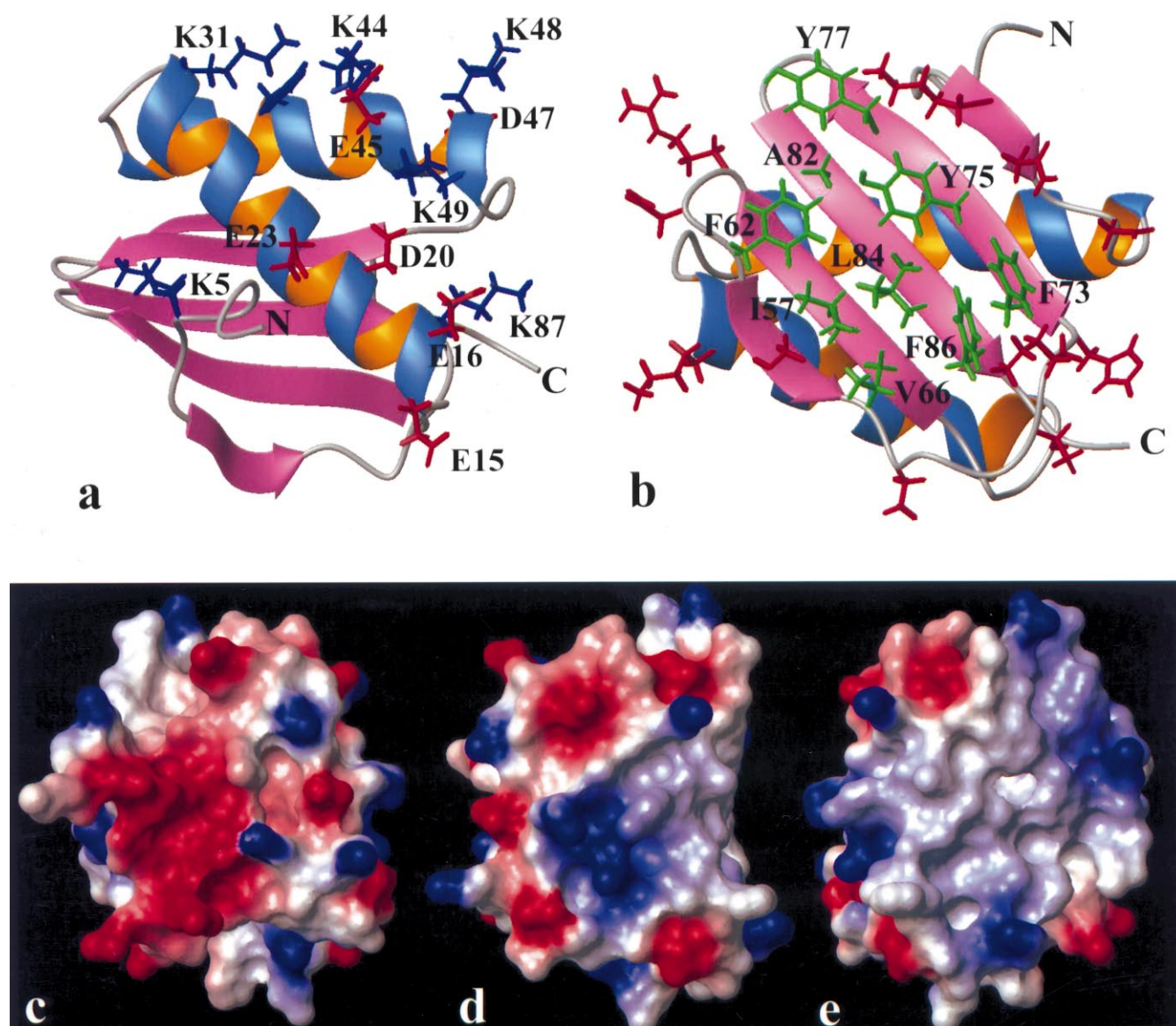


Fig. 7. Charge distribution and molecular surface of LC8. The acidic (red) and basic (blue) side chains emanating from helices $\alpha 1$ and $\alpha 2$ are shown in panel a. The opposite face (b) is formed from a series of hydrophobic side chains (green) surrounded by hydrophilic residues (both acidic and basic side chains are shown in red). Molecular surface representations (c–e) indicate that helices $\alpha 1$ and $\alpha 2$ form acidic and basic patches, respectively. The β sheet forms a concave hydrophobic pocket with residue L84 at the base. This hydrophobic patch is surrounded by charged residues that likely contribute significantly to the specificity of interactions involving this region. The coordinates (pdb accession code 1bkq) are from [89].

atic mutagenesis to probe LC8 function in vivo and in vitro.

6. The Tctex1 dynein light chain family and meiotic drive of the *t* complex

In addition to ICs and the LC8 protein, all dyneins

with multiple HCs also contain one or more members of the Tctex1 family. This diverse group of LCs was first identified in cytoplasmic dynein [90]. Subsequently, a homologous polypeptide was identified in *Chlamydomonas* inner arm I1 [91] and a more distantly related family member (termed Tctex2) was found in the outer arm [92]. Examination of the current databases has revealed several additional

members of this family including the rp3 protein that is also a demonstrable component of cytoplasmic dynein [93]; this protein was originally of interest as a candidate for involvement in X-linked retinitis pigmentosa [94]. The Tctex1 and rp3 light chains are differentially regulated in both a developmental and tissue-specific manner. For example, Tctex1 message is abundant in fetal brain but is downregulated in the adult [95] whereas message levels for the rp3 light chain reveal the opposite pattern [94]. This has led to the hypothesis that these LCs play a role in the binding of particular cargoes or in differentially regulating some essential aspect of dynein function. Furthermore, immunolocalization and biochemical fractionation studies have suggested that functionally distinct populations of cytoplasmic dynein may contain different Tctex1 family LCs [93,96].

Both Tctex1 and Tctex2 were first identified in mice where they are encoded within the *t* complex region of chromosome 17 [97,98]. The interest in this genomic segment derives from several unusual features (see [99,100] for reviews and see Table 1). Variant forms of the *t* complex (known as *t* haplotypes) contain large inversions and are thus unable to undergo recombination with wild type. Therefore, the entire 30–40 Mb region is transmitted as a single unit. Heterozygous (+/*t*) males pass the *t* haplotype form of the chromosome to essentially all of their offspring in a non-Mendelian process known as transmission ratio distortion or meiotic drive. Although there is no apparent phenotypic consequence in the heterozygous state, homozygosity for a given *t* haplotype results in embryonic lethality due

to the presence of recessive lethal factors. This extreme meiotic drive effect is thought to be due to the activity of a series of interacting distorter and responder proteins [101]. Physiologically, meiotic drive appears to be caused by defects in spermiogenesis that result in specific sperm populations being unable to fertilize an oocyte. Importantly, *t^x/t^y* hemizygotes (i.e., animals which are homozygous for the entire genome except at the *t* complex region where they carry two haplotypes with complementing recessive lethal factors) produce sperm with extremely poor motility and are completely sterile [101,102].

Tctex1 and Tctex2 are candidates for two of the distorter loci that are partly responsible for meiotic drive (it appears that three such loci of different strengths are required for the complete effect). This candidacy is based on a number of factors including gene location, aberrant expression in *t/t* testis and the presence of *t* haplotype-specific mutations that could affect protein structure and function. Identification of these proteins as integral components of flagellar dyneins has suggested a molecular mechanism by which meiotic drive might be achieved [91,92]. This model relies on the responder being involved in protein sorting at the level of the basal body during flagellar assembly, i.e., it acts to determine what proteins may enter the growing flagellum. The *t*-specific mutations in the responder are hypothesized to reduce its affinity for the mutant distorter (dynein) proteins thereby disallowing their incorporation into the growing axoneme. This would result in sperm containing the *t* haplotype nucleus being associated with a flagellar tail bearing wild-type dyneins. The corollary would be sperm with the wild-type nucleus (and with the wild-type responder at the basal body) incorporating both wild-type and mutant dyneins. This inclusion of mutant dyneins is predicted to result in abnormal motility. Indeed, heterozygous mice produce two distinct motile populations of sperm. One class shows the motility characteristics of wild-type sperm whereas the second have the extremely poor motility observed in sperm from *t^x/t^y* hemizygotes.

Intriguingly, mapping of dynein HC genes has revealed that one presumptive inner arm HC also is encoded within the *t* complex and, at the current level of resolution, is indistinguishable from the strongest distorter locus [103,104]. Thus, it is feasible

Table 1
Transmission of mouse *t* haplotypes^a

Male	Female	Progeny	Phenotype
+/ <i>t^x</i> ^b	x +/+	+/ <i>t^x</i>	Normal
+/ <i>t^x</i>	x +/ <i>t^x</i> ^c	+/ <i>t^x</i> <i>t^x/t^x</i> ^c	Normal Embryonic lethal
+/ <i>t^x</i>	x +/ <i>t^y</i>	+/ <i>t^x</i> <i>t^x/t^y</i>	Normal Male sterile

^a *t^x* and *t^y* refer to different *t* haplotypes that contain complementing embryonic lethal factors.

^b Heterozygous males pass the *t* haplotype to essentially all of their progeny.

^c Females transmit *t* haplotypes in a normal Mendelian manner.

that all the *t* complex distorter loci may encode components of flagellar dyneins. If this is indeed the case, the interesting locus now becomes the responder which must act in a cell-specific fashion to differentially partition the distorter effects to sperm bearing either the wild-type- or *t* haplotype-containing nucleus. At present there is no viable candidate for this protein.

7. The *roadblock*/LC7 light chain family

The most recent class of dynein LC reported contains closely related homologues that are present in both flagellar and cytoplasmic dyneins [105]. These proteins have been identified within the *Chlamydomonas* flagellar outer arm (LC7) and mammalian cytoplasmic dynein. Moreover, mutations in a related *Drosophila* polypeptide (termed *roadblock*) lead to defects in axonal transport and mitosis suggesting that these ~12 kDa proteins play an essential role in dynein function. Intriguingly, the *roadblock*/LC7 protein family also contains an additional *Drosophila* protein termed *bithoraxoid* (*bxd*). The *bxd* genomic region is part of the homeotic *bithorax* complex which controls posterior thoracic and anterior abdominal segmental identity [106,107]. Elements within the *bxd* region have been thought to act (in *cis*) essentially as enhancers to control levels of *ultrabithorax* expression. However, transcription from this region also appears to yield transacting factors. The ‘early’ RNAs are expressed during embryogenesis and do not have protein-coding potential. The ‘late’ RNA is transcribed during the third larval instar, in the pupa and also in the adult. This RNA is found in the abdomen but not the thorax and encodes the *bxd* protein that is related to *roadblock*/LC7. These observations have raised the possibility that the *bxd* protein is dynein-associated and that some aspects of *bithoraxoid* function during homeotic control may involve dynein.

8. Putting the pieces together: assembly of the dynein particle

Genetic studies in *Chlamydomonas* have revealed that two of the HCs (β and γ), both ICs and several

Table 2
Mutations affecting the outer dynein arm

Component	Gene	Properties and mutant phenotype
α HC	ODA11	Motor. Mutants can assemble dynein lacking this HC and LC5, and swim slightly slower than wild type
β HC	ODA4	Motor. Required for arm assembly. Truncation mutant lacking only the motor unit can assemble, but swims slowly
γ HC	ODA2	Motor. Required for arm assembly
DC1	ODA3	83 kDa docking complex protein. Required for arm assembly
IC1	ODA9	76 kDa microtubule-binding intermediate chain. Required for arm assembly
IC2	ODA6	63 kDa intermediate chain. Required for arm assembly
DC2	ODA1	62.5 kDa docking complex protein. Required for arm assembly
DC3	ODA14	25 kDa docking complex protein. Required for arm assembly.
LC1	n.k.	22 kDa leucine-rich repeat protein bound to motor domain of γ HC. Maps to linkage group II near centromere
LC2	ODA12	Tctex2 (related to Tctex1) associated with ICs and LC6 at base. Required for arm assembly
LC3	n.k.	17 kDa thioredoxin bound to β HC. Maps to linkage group XII/XIII
LC4	n.k.	18 kDa Ca^{2+} -binding EF-hand protein associated with γ HC. Maps to linkage group I near PF4
LC5	n.k.	14 kDa thioredoxin bound to α HC. Does not assemble in <i>oda11</i> . Maps to linkage group XIX
LC6	ODA13	14 kDa LC8 homologue. Interacts with LC2 at base. Maps to linkage group III
LC7	n.k.	12 kDa protein related to <i>Drosophila roadblock</i> and <i>bithoraxoid</i> . Similar LC present in cytoplasmic dynein. Maps to linkage group VIII
LC8	FLA14	Highly conserved. Found in several other enzymes. Required for outer and inner arm assembly and retrograde intraflagellar transport

(but not all) of the LCs are essential for assembly of the outer arm dynein particle [108]. Null mutants for any of these components results in a complete outer arm-less phenotype (see Table 2). Surprisingly however, this normally three-headed dynein is able to assemble in the complete absence of the α HC and its associated LC [13], supporting the idea that this third HC represents an elaboration of the basic two-

headed structural theme. Similarly, HC and IC mutations in inner arm I1 and null mutants for the p28 LC in inner arms I2/3 also result in assembly defects [109–111].

Considering the polypeptide complexity and enormous size of some of the components, it seems likely that dyneins are constructed through a precisely choreographed assembly process. For example, efforts to fractionate dynein particles into smaller units have suggested the existence of an IC/LC complex as a distinct and stable substructure (see [112] for review) and also indicate that each HC is tightly associated with at least one LC [113]. Recent analysis of flagellar dyneins isolated from *Chlamydomonas* cytoplasmic extracts has lent strong support to this view and, moreover, demonstrates that particle assembly occurs within the cytoplasm [114]. Assembly is initiated through the association of the two ICs and the subsequent binding of the HCs. Several LCs and the outer arm dynein docking complex are thought to also preassemble into distinct particles before incorporation into the nascent dynein. Failure to produce any of these components therefore can result in only partial assembly products that are not competent for transport into the flagellum. Indeed the proteins within partially assembled dyneins appear to have significantly reduced stability in the cytoplasm [114].

9. Conclusions

Recent studies have started to shed light on the nature of the bewildering array of proteins that comprise various dynein particles. Current models for the general organization of outer arm, inner arm I1 and cytoplasmic dyneins suggest that these particles contain HCs, several ICs involved in cargo attachment, and at least three distinct classes of LC (namely the highly conserved LC8, and members of the *roadblock*/LC7 and Tctex1 protein families) which are associated with the ICs at the base. This clear similarity in form strongly supports the hypothesis that these three dynein types originated from a single two-headed ancestral dynein motor. The remaining inner arms (arms I2/3) consist of only single HCs associated with actin, p28 and centrin, and thus appear to represent a distinct dynein subclass.

Clearly, much has been learnt about dynein in the

last few years. Some of the challenges ahead include determining (i) the mechanisms by which cargo attachment is achieved and regulated, (ii) the role of the generic light chain classes, (iii) how individual HCs are regulated, (iv) the mechanism of force generation, and (v) why some dyneins contain multiple motor domains with distinct motility properties. Finally, a major long-term goal will be to provide a structural basis for understanding these various aspects of dynein function.

10. Note added in proof

Recently, the X-ray crystal structure of the LC8 dimer in complex with a 12-residue peptide from neuronal nitric oxide synthase (nNOS) has been solved (J. Liang, S.R. Jaffrey, S.H. Snyder, J. Clardy, *Nat. Struct. Biol.* 6 (1999) 735–740). This study confirms that LC8 is indeed dimeric and demonstrates that two identical nNOS binding sites are formed adjacent to the interface between the two monomers.

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